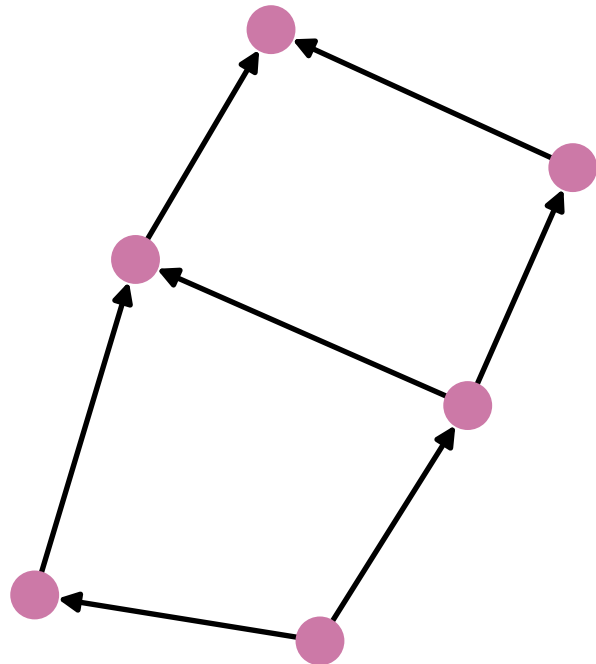
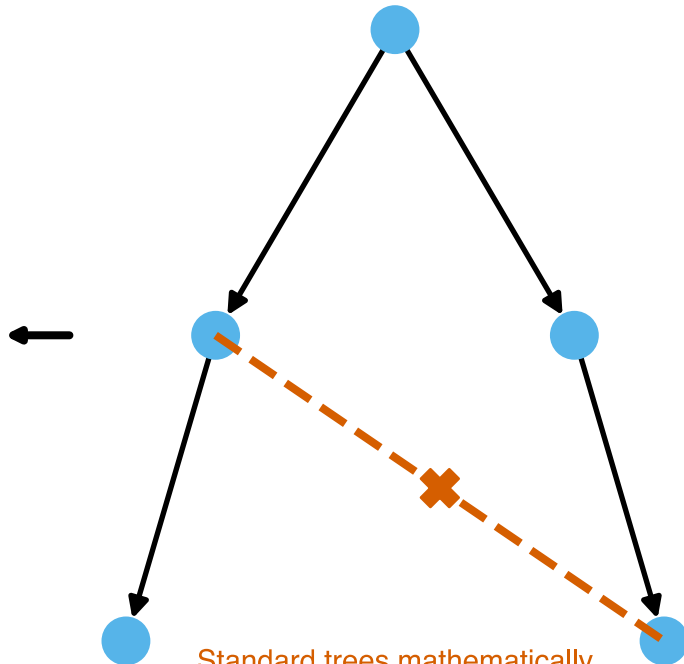


Biological Reality
(Reticulate Evolution)



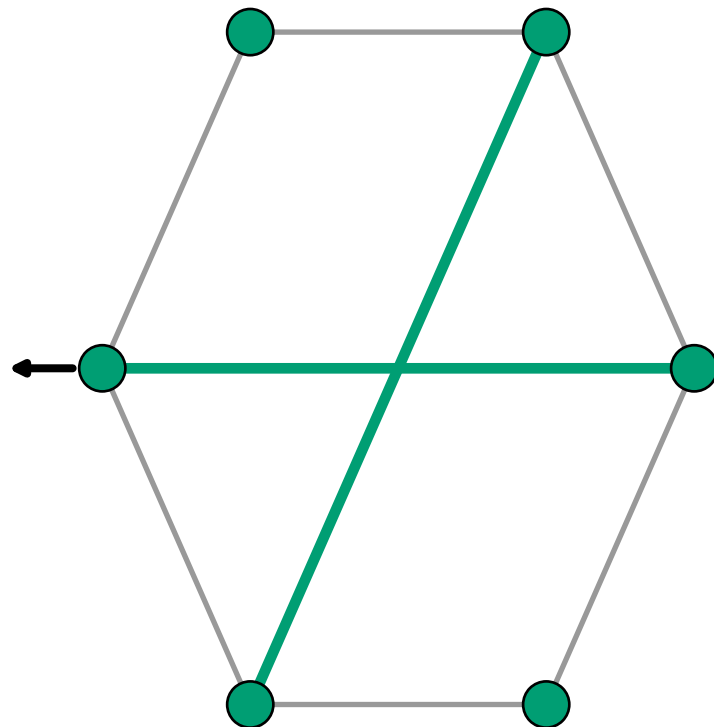
Horizontal Gene Transfer
creates web-like structures

Traditional Phylogeny
(Forced Bifurcation)



Standard trees mathematically
break topological loops

ATLAZ Geometric Engine
(Topological Resolution)



$\text{GF}(2)$ matrix reduction directly
computes exact H_1 persistence